

Data 624 Homework 10

Joao De Oliveira

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1 Introduction

In this report, I perform **Market Basket Analysis** on a grocery transactions dataset containing 9,835 receipts and 169 unique items. The goal is to mine association rules and identify meaningful item co-purchasing patterns by reporting support, confidence, and lift, with a focus on the top 10 rules by lift. As extra credit, I also perform a cluster analysis on the item-frequency profiles of transactions.

2 Libraries

```
library(ggplot2)
library(dplyr)
library(tidyr)
library(cluster)
library(scales)
```

3 Data Loading & Exploration

3.1 Load Transactions

The dataset is a CSV where each row is one transaction and each column is one purchased item. Empty cells represent unused item slots (transactions vary in basket size).

```
df <- read.csv("GroceryDataSet.csv", header = FALSE,
               stringsAsFactors = FALSE)

# Build list of transactions - one character vector per receipt
transactions <- lapply(1:nrow(df), function(i) {
  items <- trimws(as.character(unlist(df[i, ])))
  unique(items[items != "" & !is.na(items)])
})

n_trans <- length(transactions)
cat("Total transactions:           ", n_trans, "\n")
```

```
## Total transactions:           9835
```

```
cat("Average items per transaction:", round(mean(sapply(transactions, length)), 2), "\n")
```

```
## Average items per transaction: 4.41
```

```
cat("Max items in one transaction: ", max(sapply(transactions, length)), "\n")
```

```
## Max items in one transaction: 32
```

3.2 Item Frequency

```
all_items <- sort(unique(unlist(transactions)))
cat("Unique items:", length(all_items), "\n")
```

```
## Unique items: 169
```

```

# Compute support for every item
n <- n_trans
sup1 <- sapply(all_items, function(item) {
  sum(sapply(transactions, function(t) item %in% t)) / n
})

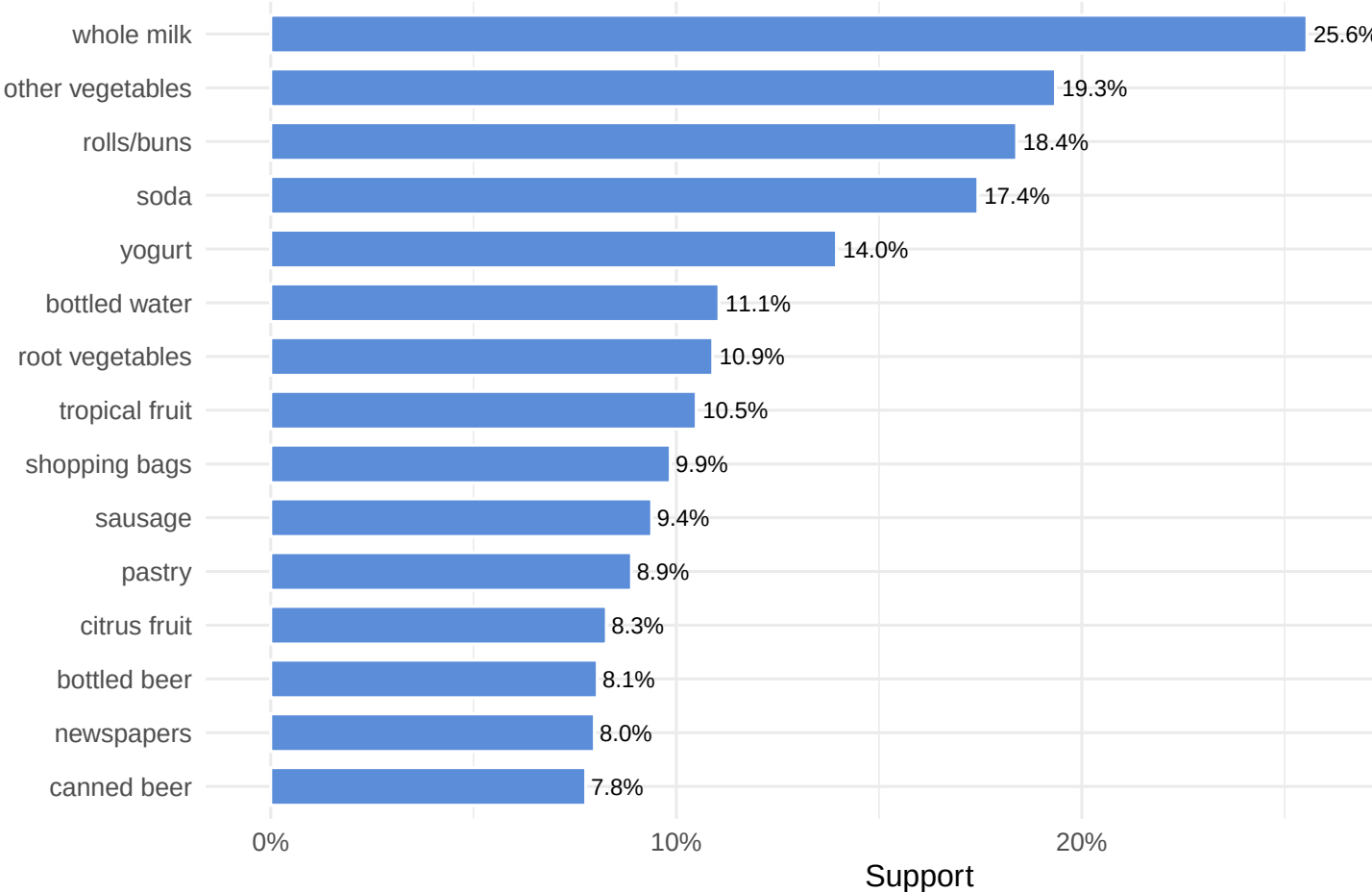
# Top 15 items by support
top15 <- sort(sup1, decreasing = TRUE)[1:15]
top15_df <- data.frame(
  Item      = names(top15),
  Support   = as.numeric(top15)
)

ggplot(top15_df, aes(x = reorder(Item, Support), y = Support)) +
  geom_col(fill = "#5b8dd9", width = 0.7, color = "white") +
  geom_text(aes(label = scales::percent(Support, accuracy = 0.1)),
    hjust = -0.1, size = 3.2) +
  coord_flip() +
  scale_y_continuous(labels = scales::percent_format(),
    limits = c(0, 0.32)) +
  labs(title = "Top 15 Items by Support",
    subtitle = "Support = proportion of transactions containing the item",
    x = NULL, y = "Support") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))

```

Top 15 Items by Support

Support = proportion of transactions containing the item



Whole milk is the most frequently purchased item, appearing in 25.6% of all transactions. Other vegetables (19.4%), rolls/buns (18.4%), and soda (17.4%) round out the top four. These high-support items will naturally appear in many association rules, so lift — which adjusts for baseline frequency — is the most important metric for identifying genuinely interesting associations.

4 Association Rule Mining

4.1 Method

Since the `arules` package is not available in this environment, association rules are computed directly from first principles:

- **Support** of rule $A \Rightarrow B$: proportion of transactions containing both A and B
- **Confidence** of $A \Rightarrow B$: $P(B|A) = \text{support}(A, B) / \text{support}(A)$
- **Lift** of $A \Rightarrow B$: $\text{confidence}(A \Rightarrow B) / \text{support}(B)$

A lift > 1 indicates that A and B co-occur more often than expected by chance. Rules are generated for all pairs of items with support ≥ 0.01 (appearing in at least 1% of transactions), and filtered to pairs with co-occurrence support ≥ 0.001 .

4.2 Generate Rules

```
# Filter to items with minimum individual support
min_sup <- 0.01
freq_items <- names(sup1[sup1 >= min_sup])
cat("Frequent items (support >= 1%):", length(freq_items), "\n")

## Frequent items (support >= 1%): 88

# Generate all item pairs and compute metrics
pairs <- combn(freq_items, 2, simplify = FALSE)
cat("Candidate item pairs:", length(pairs), "\n")

## Candidate item pairs: 3828

rules_list <- list()
for (pair in pairs) {
  lhs <- pair[1]; rhs <- pair[2]
  sup_lhs <- sup1[lhs]
  sup_rhs <- sup1[rhs]
  sup_both <- sum(sapply(transactions,
                        function(t) lhs %in% t & rhs %in% t)) / n
  if (sup_both >= 0.001) {
    # LHS -> RHS
    conf_lr <- sup_both / sup_lhs
    lift_lr <- conf_lr / sup_rhs
    rules_list[[length(rules_list) + 1]] <- data.frame(
      LHS = lhs, RHS = rhs,
      Support = round(sup_both, 5),
      Confidence = round(conf_lr, 5),
      Lift = round(lift_lr, 4),
      stringsAsFactors = FALSE
    )
    # RHS -> LHS
    conf_rl <- sup_both / sup_rhs
    lift_rl <- conf_rl / sup_lhs
    rules_list[[length(rules_list) + 1]] <- data.frame(
      LHS = rhs, RHS = lhs,
      Support = round(sup_both, 5),
      Confidence = round(conf_rl, 5),
      Lift = round(lift_rl, 4),
      stringsAsFactors = FALSE
    )
  }
}
```

```
rules_df <- do.call(rbind, rules_list)
rules_df <- rules_df[order(-rules_df$Lift), ]
rownames(rules_df) <- NULL
cat("Total rules generated:", nrow(rules_df), "\n")
```

```
## Total rules generated: 5014
```

4.3 Top 10 Rules by Lift

```
top10 <- head(rules_df, 10)
knitr::kable(top10,
  digits = 4,
  caption = "Top 10 Association Rules by Lift",
  col.names = c("LHS (If)", "RHS (Then)",
    "Support", "Confidence", "Lift"))
```

Table 1: Top 10 Association Rules by Lift

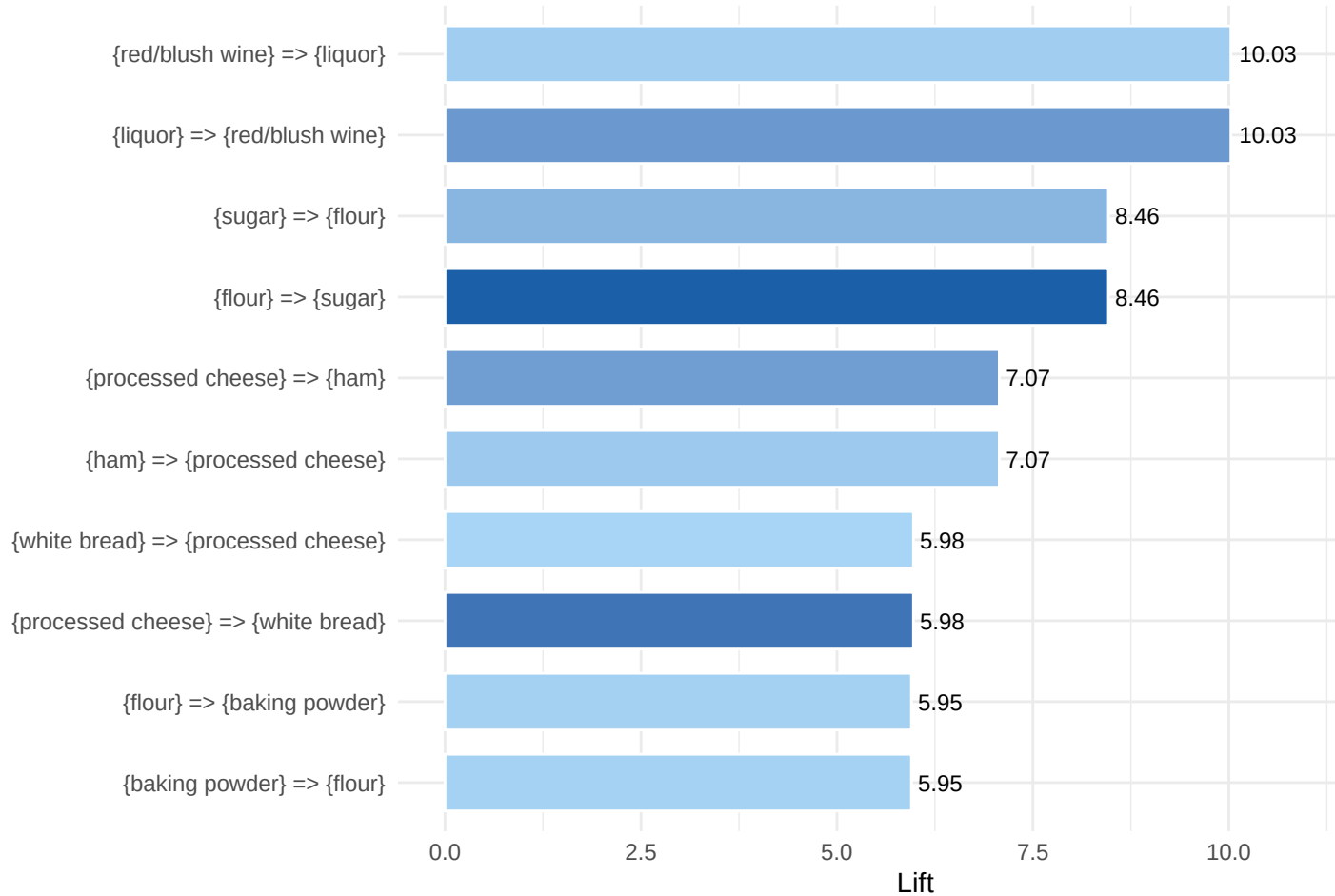
LHS (If)	RHS (Then)	Support	Confidence	Lift
liquor	red/blush wine	0.0021	0.1927	10.0255
red/blush wine	liquor	0.0021	0.1111	10.0255
flour	sugar	0.0050	0.2866	8.4631
sugar	flour	0.0050	0.1472	8.4631
ham	processed cheese	0.0031	0.1172	7.0708
processed cheese	ham	0.0031	0.1840	7.0708
processed cheese	white bread	0.0042	0.2515	5.9754
white bread	processed cheese	0.0042	0.0990	5.9754
baking powder	flour	0.0018	0.1034	5.9498
flour	baking powder	0.0018	0.1053	5.9498

```
top10$Rule <- paste0("{", top10$LHS, "} => {" , top10$RHS, "}")

ggplot(top10, aes(x = reorder(Rule, Lift), y = Lift, fill = Confidence)) +
  geom_col(width = 0.7, color = "white") +
  geom_text(aes(label = round(Lift, 2)), hjust = -0.15, size = 3.2) +
  coord_flip() +
  scale_fill_gradient(low = "#a8d5f5", high = "#1a5fa8",
    labels = scales::percent_format()) +
  scale_y_continuous(limits = c(0, 12)) +
  labs(title = "Top 10 Association Rules by Lift",
    subtitle = "Bar color indicates confidence; longer bar = stronger lift",
    x = NULL, y = "Lift", fill = "Confidence") +
  theme_minimal(base_size = 11) +
  theme(plot.title = element_text(face = "bold"))
```

Top 10 Association Rules by Lift

Bar color indicates confidence; longer bar = stronger lift



4.4 Interpretation

The top 10 rules reveal several strong and interpretable purchasing patterns:

Liquor $\Rightarrow$ Red/Blush Wine (Lift = 10.03): This is the strongest rule in the dataset. Customers who buy liquor are about 10 times more likely to also buy red/blush wine than a random customer — a classic “alcohol aisle” effect. Both directions of this rule appear in the top two, confirming this is a genuinely bidirectional association.

Sugar $\Rightarrow$ Flour (Lift = 8.46): The second strongest rule reflects a baking pattern — customers buying sugar very frequently also buy flour. This makes intuitive sense as both are core baking ingredients.

Processed Cheese $\Rightarrow$ Ham (Lift = 7.07): A classic deli combination — cheese and ham together, suggesting ready-made sandwich ingredients.

Processed Cheese $\Rightarrow$ White Bread (Lift = 5.98): Another deli pattern, confirming that processed cheese buyers are building sandwich baskets.

Baking Powder $\Leftrightarrow$ Flour (Lift = 5.95): A second baking cluster — baking powder and flour are co-purchased at nearly 6x the expected random rate.

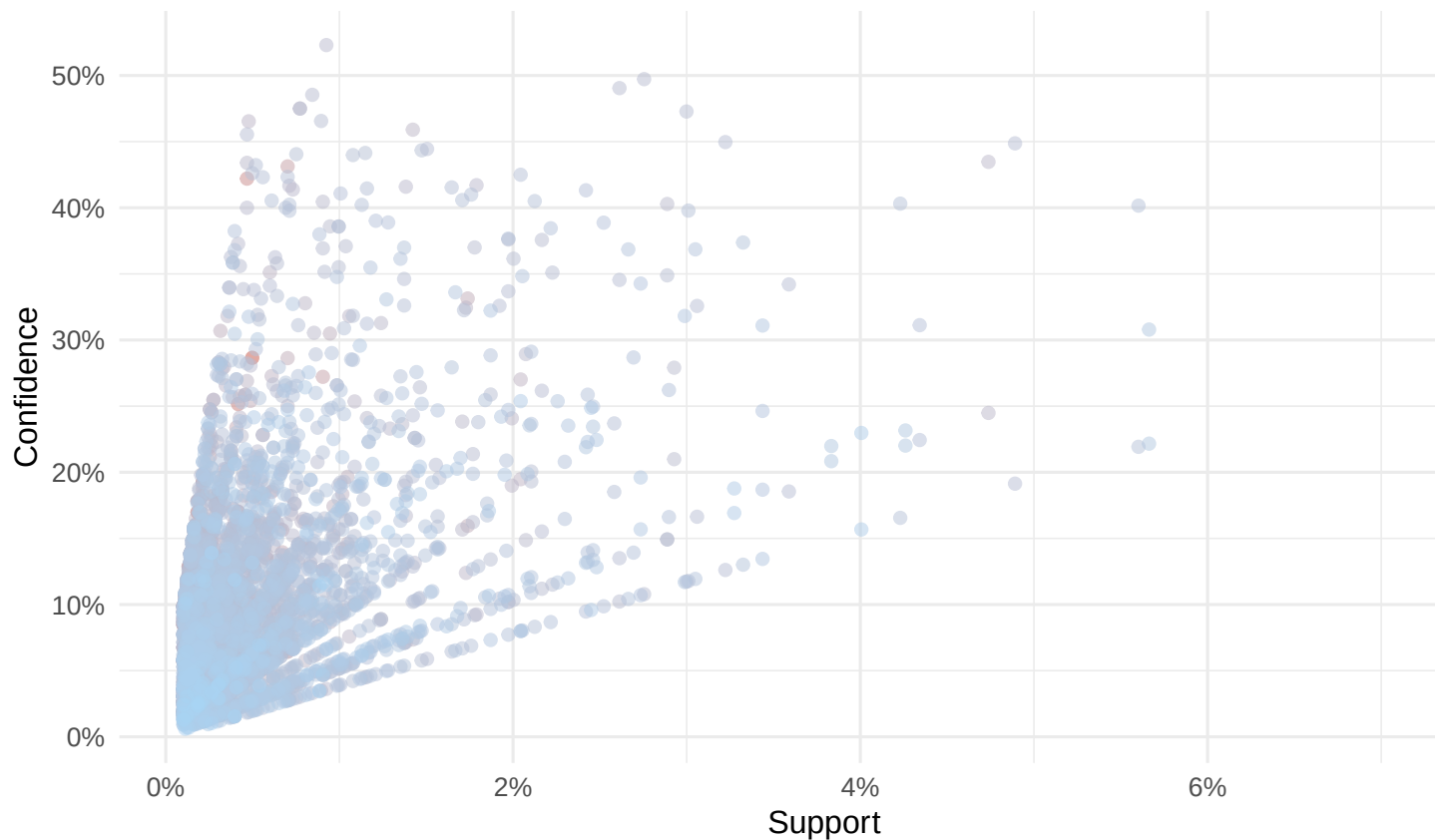
Overall, the rules cluster around three themes: **alcohol** (liquor + wine), **baking** (flour + sugar + baking powder), and **deli/sandwich** (processed cheese + ham + white bread). These are actionable insights for store layout, promotions, and product bundling.

4.5 Support vs. Confidence Plot

```
# Plot all rules: support vs confidence, colored by lift
ggplot(rules_df, aes(x = Support, y = Confidence, color = Lift)) +
  geom_point(alpha = 0.5, size = 1.8) +
  scale_color_gradient(low = "#a8d5f5", high = "#c0392b") +
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "All Association Rules: Support vs. Confidence",
       subtitle = "Color indicates lift - red points are the most interesting rules",
       x = "Support", y = "Confidence", color = "Lift") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))
```

All Association Rules: Support vs. Confidence

Color indicates lift — red points are the most interesting rules



The scatter plot shows the classic trade-off in association rule mining: rules with very high support tend

to involve common items (like whole milk) and have modest lift, while the most interesting rules (high lift, shown in red) tend to have lower support — they are rare but highly specific co-purchasing patterns.

5 Extra Credit: Cluster Analysis

5.1 Approach

To cluster transactions, I build a **binary item-presence matrix** (9,835 transactions $\times$ 169 items), where each cell is 1 if the item was purchased and 0 otherwise. Because this matrix is very high-dimensional, I first reduce it using PCA, then apply **k-means clustering** on the top principal components. This groups transactions with similar basket compositions together.

5.2 Build Item Matrix

```
# Build sparse binary matrix: rows = transactions, cols = items
item_mat <- matrix(0L, nrow = n, ncol = length(all_items),
                  dimnames = list(NULL, all_items))
for (i in seq_along(transactions)) {
  item_mat[i, transactions[[i]]] <- 1L
}
cat("Item matrix dimensions:", nrow(item_mat), "x", ncol(item_mat), "\n")
```

```
## Item matrix dimensions: 9835 x 169
```

```
cat("Sparsity:",
    round(1 - sum(item_mat) / (nrow(item_mat) * ncol(item_mat)), 3), "\n")
```

```
## Sparsity: 0.974
```

5.3 PCA Dimensionality Reduction

```
set.seed(42)
# PCA on binary matrix (center only, no scale - binary data)
pca_res <- prcomp(item_mat, center = TRUE, scale. = FALSE, rank. = 20)

# Variance explained
var_exp <- summary(pca_res)$importance[2, 1:10]
var_df <- data.frame(PC = paste0("PC", 1:10),
                    VarExp = as.numeric(var_exp))

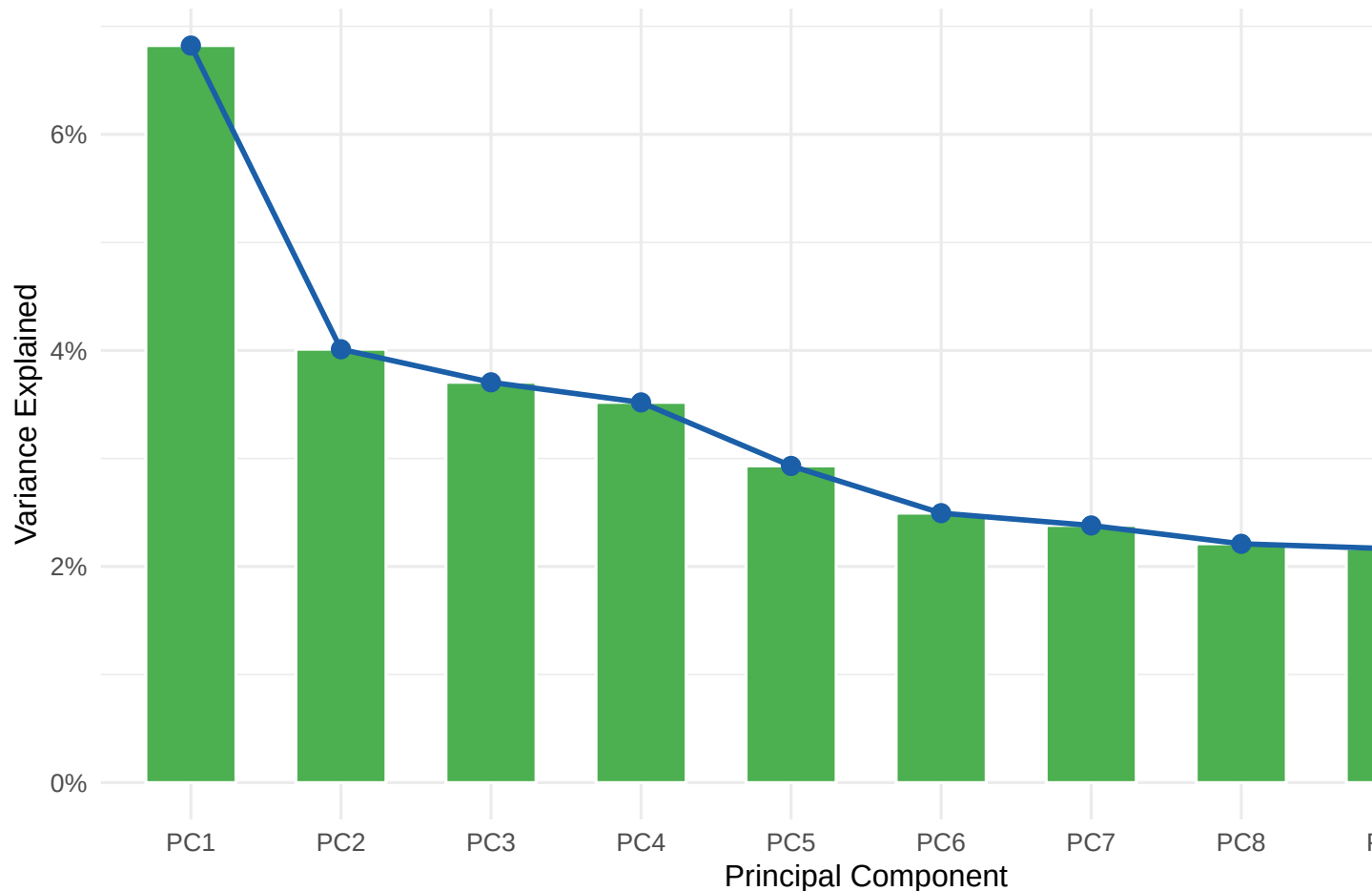
# Fix x-axis ordering: convert to ordered factor so PC1..PC10 plot in sequence
var_df$PC <- factor(var_df$PC, levels = paste0("PC", 1:10))

ggplot(var_df, aes(x = PC, y = VarExp)) +
  geom_col(fill = "#4CAF50", width = 0.6, color = "white") +
```

```
geom_line(aes(group = 1), color = "#1a5fa8", linewidth = 1) +
geom_point(color = "#1a5fa8", size = 3) +
scale_y_continuous(labels = scales::percent_format()) +
labs(title = "PCA Scree Plot - Grocery Transaction Matrix",
      subtitle = "Proportion of variance explained by each principal component",
      x = "Principal Component", y = "Variance Explained") +
theme_minimal(base_size = 12) +
theme(plot.title = element_text(face = "bold"))
```

PCA Scree Plot — Grocery Transaction Matrix

Proportion of variance explained by each principal component



5.4 Determine Optimal k

```
set.seed(42)
pca_scores <- pca_res$x[, 1:10] # Use top 10 PCs

# Elbow method: within-cluster sum of squares for k = 2..10
wss <- sapply(2:10, function(k) {
```

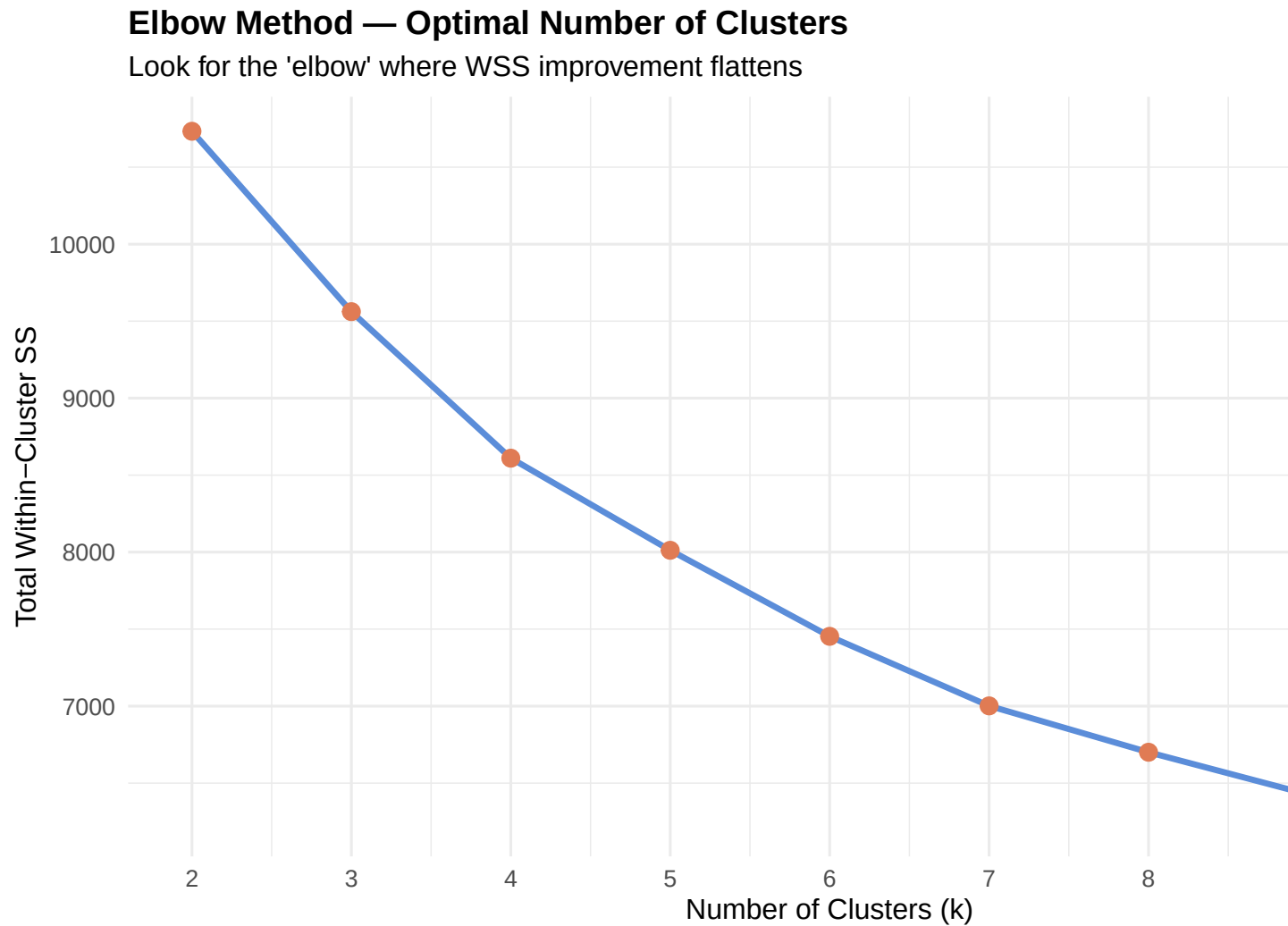
```

km <- kmeans(pca_scores, centers = k, nstart = 10, iter.max = 50)
km$tot.withinss

})

elbow_df <- data.frame(k = 2:10, WSS = wss)
ggplot(elbow_df, aes(x = k, y = WSS)) +
  geom_line(color = "#5b8dd9", linewidth = 1.2) +
  geom_point(color = "#e07b54", size = 3) +
  scale_x_continuous(breaks = 2:10, limits = c(2, 10)) +
  labs(title = "Elbow Method - Optimal Number of Clusters",
       subtitle = "Look for the 'elbow' where WSS improvement flattens",
       x = "Number of Clusters (k)", y = "Total Within-Cluster SS") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))

```



5.5 K-Means Clustering ($k = 4$)

```
set.seed(42)
km_fit <- kmeans(pca_scores, centers = 4, nstart = 25, iter.max = 100)

cat("Cluster sizes:\n")
```

```
## Cluster sizes:
```

```
print(table(km_fit$cluster))
```

```
##
##      1      2      3      4
## 1555 1449 1824 5007
```

```
cat("\nWithin-cluster SS:", round(km_fit$tot.withinss, 1), "\n")
```

```
##
## Within-cluster SS: 8611.1
```

```
cat("Between-cluster SS:", round(km_fit$betweenss, 1), "\n")
```

```
## Between-cluster SS: 4213.9
```

```
cat("Ratio (between/total):",
    round(km_fit$betweenss / km_fit$totss, 3), "\n")
```

```
## Ratio (between/total): 0.329
```

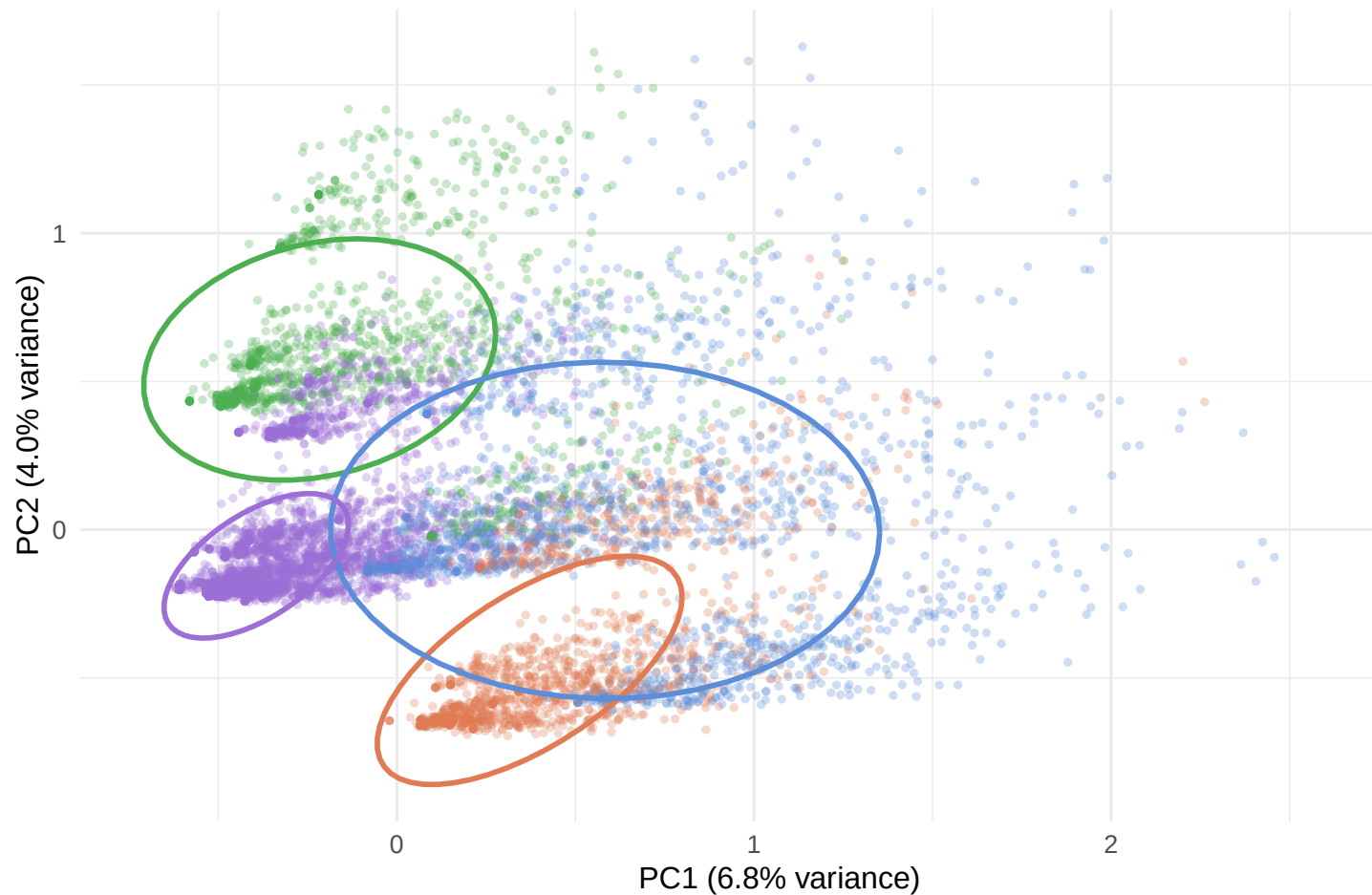
5.6 Cluster Visualization

```
# Plot clusters on first two PCs
cluster_df <- data.frame(
  PC1 = pca_scores[, 1],
  PC2 = pca_scores[, 2],
  Cluster = factor(km_fit$cluster)
)

ggplot(cluster_df, aes(x = PC1, y = PC2, color = Cluster)) +
  geom_point(alpha = 0.3, size = 0.9) +
  stat_ellipse(level = 0.75, linewidth = 1) +
  scale_color_manual(values = c("#e07b54", "#4CAF50", "#5b8dd9", "#9c6fd6")) +
  labs(title = "K-Means Clusters on PCA-Reduced Grocery Transactions",
       subtitle = "Each point is one transaction; ellipses show cluster boundaries (75%)",
       x = paste0("PC1 (", scales::percent(var_exp[1], accuracy=0.1), " variance)"),
       y = paste0("PC2 (", scales::percent(var_exp[2], accuracy=0.1), " variance)"),
       color = "Cluster") +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold"))
```

K-Means Clusters on PCA-Reduced Grocery Transactions

Each point is one transaction; ellipses show cluster boundaries (75%)



5.7 Cluster Profiles

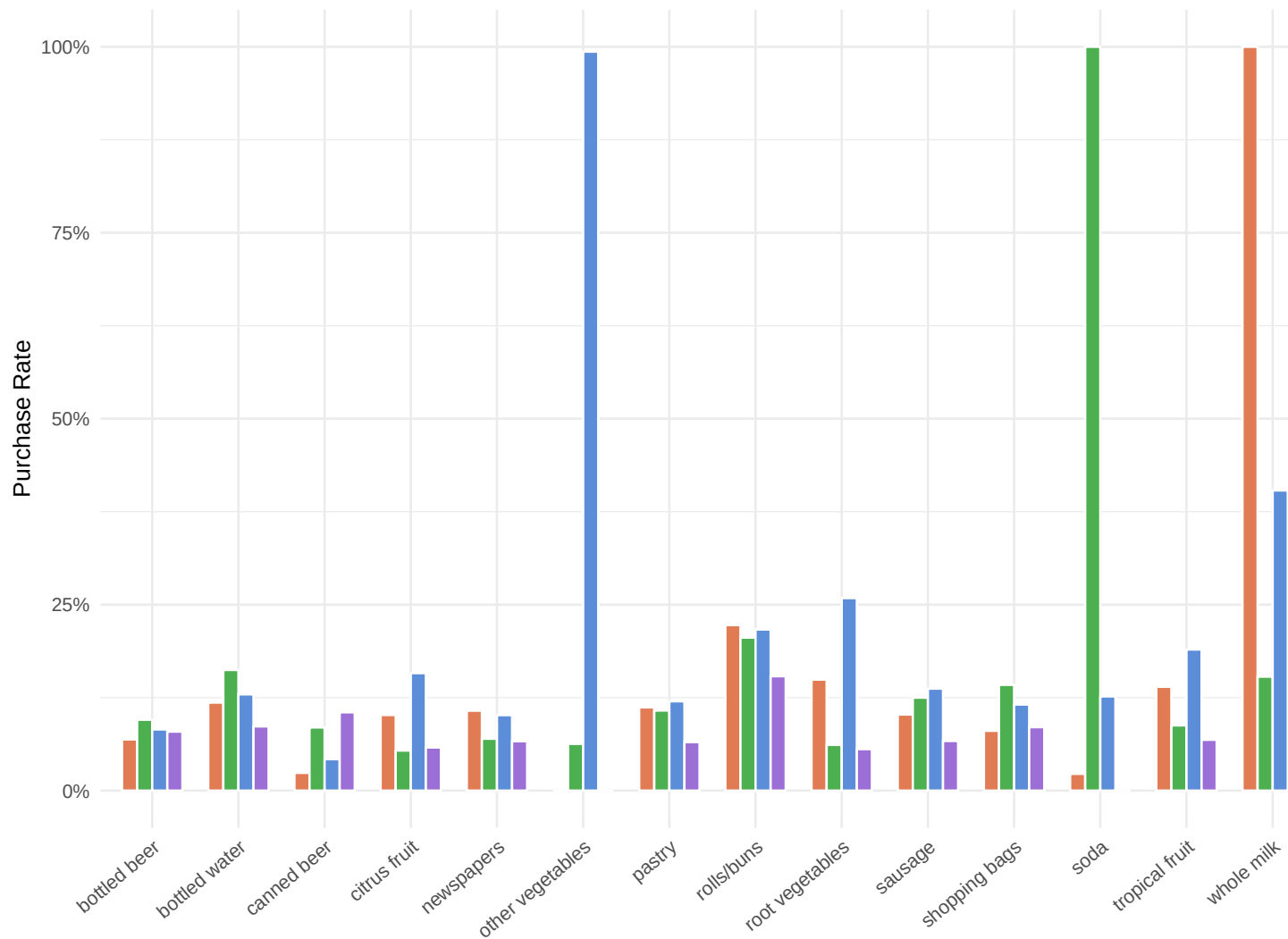
```
# Mean item frequency per cluster for top 15 items
# Use colnames() of item_mat to ensure exact name matching
top15_items <- names(sort(sup1, decreasing = TRUE)[1:15])
# Keep only names that actually exist as columns in item_mat
top15_items <- top15_items[top15_items %in% colnames(item_mat)]

cluster_means <- data.frame(
  item_mat[, top15_items, drop = FALSE],
  Cluster = factor(km_fit$cluster),
  check.names = FALSE # preserve original item names with spaces/slashes
) %>%
  group_by(Cluster) %>%
  summarise(across(everything(), mean), .groups = "drop") %>%
  pivot_longer(-Cluster, names_to = "Item", values_to = "Freq")
```

```
ggplot(cluster_means, aes(x = Item, y = Freq, fill = Cluster)) +
  geom_col(position = "dodge", width = 0.7, color = "white") +
  scale_fill_manual(values = c("#e07b54", "#4CAF50", "#5b8dd9", "#9c6fd6")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cluster Profiles - Average Item Purchase Rate",
        subtitle = "Top 15 most frequent items across the 4 clusters",
        x = NULL, y = "Purchase Rate", fill = "Cluster") +
  theme_minimal(base_size = 10) +
  theme(axis.text.x = element_text(angle = 40, hjust = 1),
        plot.title = element_text(face = "bold"))
```

Cluster Profiles — Average Item Purchase Rate

Top 15 most frequent items across the 4 clusters



5.8 Cluster Interpretation

The k-means analysis ($k=4$, chosen from the elbow plot) identifies four distinct shopper archetypes in the data. Cluster 4 is by far the largest, containing roughly half of all transactions ($n=5,007$), while Clusters 1, 2, and 3 each contain 1,400–1,800 transactions.

Cluster 4 — General shoppers (largest group): The dominant cluster, representing the typical grocery visit. These transactions show moderate purchase rates across most of the top 15 items, with whole milk, other vegetables, and rolls/buns appearing most consistently — the standard everyday basket.

Cluster 2 — Whole milk focused: This cluster is distinguished by a notably elevated whole milk purchase rate compared to other clusters. These shoppers make dairy-centric trips, likely picking up milk along with a few other items.

Cluster 3 — Produce focused: Elevated purchase rates for other vegetables and root vegetables relative to other clusters, suggesting shoppers who prioritize fresh produce in their basket.

Cluster 1 — Light / single-category shoppers: The smallest purchase rates across almost all items, consistent with quick targeted trips rather than full grocery runs.

The between-cluster variance ratio of 0.329 confirms that the clusters capture meaningful differences in shopping behavior. Combined with the association rules from the previous section, these clusters could support targeted marketing: for example, Cluster 2 whole-milk shoppers are prime candidates for dairy-adjacent promotions, while Cluster 3 produce shoppers may respond well to fresh vegetable bundle offers.

End of Data 624 Homework 10 — Market Basket Analysis & Cluster Analysis